

Decentrum

Project Team

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Chapter 1

Introduction

The extensive use of social media apps have changed our way of communication, but some people believe that these changes are not necessarily constructive [1]. These concerns are especially due to centralized authorities in social media networks. These platforms are managed by large corporation that gathers and manipulates data of millions or billions of users. That leads to problem for data protection and privacy issues. Decentralized social media networks leverage peer-to-peer, blockchain technologies to enable users to secure their personal data and also allowing them to monetize their content[2]. **Decentrum** is a decentralized social media platform that leverage blockchain technology to provide users control over their data, by enhancing privacy to secure users data from centralized authorities and security to protect their personal information for others. [?].

1.1 Problem Statement

Centralized social media apps like facebook, Instagram fed onto the data of their users and they misuse their data for their own goals and objectives. There is also less user control, user can't post whatever they want as centralized authorities censor their content if they feel that it is not according to their guidelines. Centralized social media apps also monitor their user activities and launch their advertisements according these insights[3]. Users have have to trust a third-party to ensure that their data is protected and accessible[4].

1.2 Scope

"Decentrum's" main objective is to create a decentralized social media network where users can engage, communicate, and share any type of content freely without the intervention of any centralized authority or third-party. The platform will make use of Blockchain

technology in order to guarantee user data, privacy and transparency. The platform aims to create an environment where users decide what content they wish to see or engage with, without any interference from the centralized authority. The main functionalities include content creation where users can create the content they like and it cannot be censored unless it does not promote negativity. To prevent influx of bots on the platform, we will implement KYC mechanism. The platform also offers content monetization feature. It is up to the users to decide whether or not to watch advertisements, and those that do so will receive tokens as payment. The users will be able to engage with their family and friends through groups and community created by the users. All features will be available to the users without any premium version.

1.3 Modules

1.3.1 Content Management

In this module users shall be able to share and create content of any type. They shall be able to react on posted content.

1. Content Creation

Content Creation allow users to create or post whatever they want to it can be in the form of text or video. They can also post images or share links of article or blogs. They can also react on it through likes and comments.

2. Content Sharing

Content Sharing allow users to share content that is posted on the platform to their family and friends that can be in the form of text, video or images. They can tag their friends on any post they share.

1.3.2 Profile Management

This module shall be able to allow users to manage their profile and account.

1. Personal Info Management

This feature allow users to edit their personal information. Users can also change their cover or profile photo. They can also maintain their timeline.

2. Account & Privacy controls

User shall be maintain their account through this feature such as changing passwords. They can also manage who can see their post and vice versa. They can also customize how notifications will be retrieved.

1.3.3 Monetization and Rewards

Users shall be able to monetize their content and also be able to earn rewards.

1. Free Monetization

No premium version will be required for content creators to monetize their content. They can monetize their content for free.

2. Rewards

When a user creates a new account on this platform he will receive some rewards token to manage his transactions. Users can also earn rewards if they opt to watch ads.

1.3.4 Community Management & Interactions

This module allows users to create groups and communities to engage with their loved ones.

1. Groups

Users can form groups with their family and friends where they can share their feelings and interact by having conversations with each other in a more private manner.

2. Communities

User can notify other users about the upcoming events through making communities in the platform. They talk about certain topic to get different opinions on it.

3. Chat with Friends

Users can also engage with friends by having chat with them. They can message them privately, if they want to discuss a private matter.

1.4 User Classes and Characteristics

User class	Description
General Users	A These type of users made up most of platform population, they interact with many features which comprises of content sharing , content creation and reacting on posts that includes videos, text or images. They manage their profiles by updating their profiles and privacy settings. They also engage with others through chats, by joining communities and having group discussions.
Content Creators	These users focus on producing original content, by posting videos,images, text posts to engage with their audience. They also monetize their content to earn income through engagement with their created content. They are the key players in contributing towards platform ecosystem.
Community Builders	These users are responsible for creating and managing groups and communities, promoting discussions on particular topics. They encourage other users to participate in conversations and manage group activities. Community Builders help facilitate interactions between public and private discussions to maintain an engaged user base.
Privacy Conscious Users	These users have priority for their security and control over their communications on the platform. They are very cautious over controlling who can or can't see their posts by adjusting their privacy settings. They are picky about the content they share and even opt for customized notification settings to ensure a safe and private experience on the site.
Reward Seekers	Reward seekers are inspired by the token based reward system of the platform. To make transaction on the platforms, they engage by watching ads and other incentive based tasks to earn coins. There main goal is just to maximize their rewards, by taking advantage of built-in incentives for active interactions.
Moderators	They are responsible for maintaining the safety of platforms and to ensure that content fulfills the community guidelines. They take action on the content that is reported and take action according to the rules. Their role is so crucial in curating a healthy environment where users can feel safe while at liberty to exercise the self within the limits of what the platform allows.

Table 1.1: User Classes and Characteristics.

Chapter 2

Project Requirements

2.1 Use-case

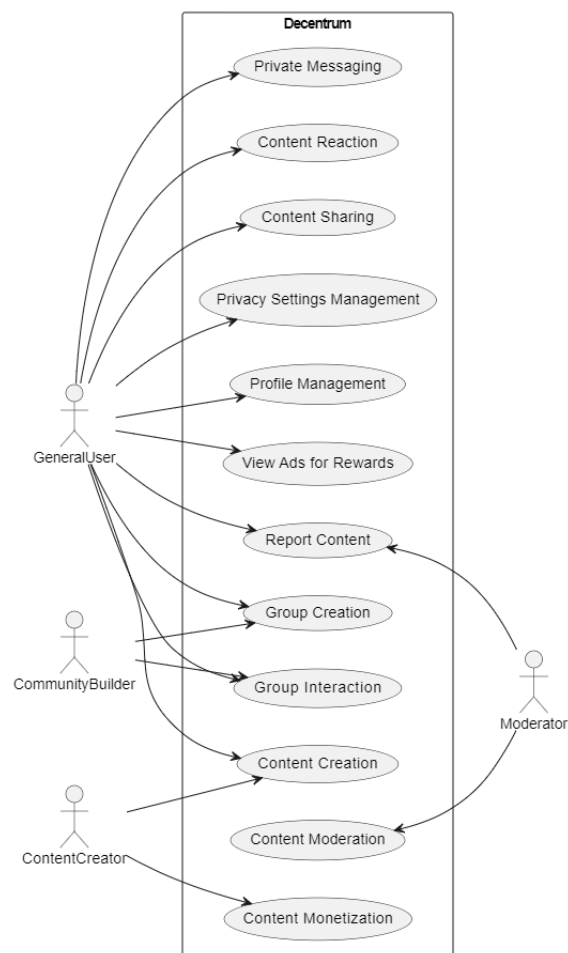


Figure 2.1: Use-Case Diagram.

Use Case UC1:	Content Creation
Actor	General User, Content Creator
Preconditions	User shall be logged in to the system.
Descriptions	User clicks on "Create Post" Option. System will displays interface for content creation. User shall create a post and clicks on submit button.

Table 2.1: Use-case UC1.

Use Case UC2:	Content Reaction
Actor	General User
Preconditions	A post shall be available on user timeline.
Descriptions	User views a post. User reacts on post. Reaction is registered and updates the reaction count.

Table 2.2: Use-case UC2.

Use Case UC3:	Content Sharing
Actor	General User
Preconditions	A post shall be available on user timeline to share.
Descriptions	User views a post. User clicks on share post button. User tags a friend in shared content. The shared posts in shared to user's timeline by system and tags him.

Table 2.3: Use-case UC3.

Use Case UC4:	Privacy Settings Management
Actor	General User
Preconditions	User shall be logged in to the system.
Descriptions	User access to the privacy settings. User changes post visibility to his preferred settings. System successfully updates privacy settings.

Table 2.4: Use-case UC4.

Use Case UC5:	Profile Management
Actor	General User
Preconditions	User shall be logged in to the system.
Descriptions	User navigates to the profile settings. User updates his/her profile. Changes made are saved.

Table 2.5: Use-case UC5.

Use Case UC6:	Group Creation
Actor	General User, Community Builder
Preconditions	User shall be logged in to the system.
Descriptions	User navigates to the creates group option and clicks it. User adds his friends or family members to the group. The group is created by the system and selected friends are invited.

Table 2.6: Use-case UC6.

Use Case UC7:	Group Interaction
Actor	General User, Community Builder
Preconditions	A group with some members shall exist.
Descriptions	Users post some kind of message in the group. The group are notified about some activity by the system.

Table 2.7: Use-case UC7.

Use Case UC8:	Content Monetization
Actor	Content Creator
Preconditions	A post has gained some popularity.
Descriptions	The post is enabled for monetization by user. Monetization request is approved by system. System tracks interaction for rewards.

Table 2.8: Use-case UC8.

Use Case UC9:	View ads for rewards
Actor	General User
Preconditions	User is already logged in.
Descriptions	User chooses to view an ad to earn tokens. System plays the ad. System credits the user's account with tokens post-viewing.

Table 2.9: Use-case UC9.

Use Case UC10:	Report Content
Actor	General User
Preconditions	The user observes inappropriate content.
Descriptions	User reports the contents. System logs the report. System sends report to moderators for review.

Table 2.10: Use-case UC10.

Use Case UC11:	Review Reported Content
Actor	Moderators
Preconditions	Content has been reported.
Descriptions	Moderators reviews the content that has been re-reported. Moderators decides to approve or remove the content.

Table 2.11: Use-case UC11.

Use Case UC12:	Private Messaging
Actor	General User
Preconditions	The user is logged in.
Descriptions	A user starts a direct conversation with a friend. The system opened a messaging window. It sends and receives the message in real-time

Table 2.12: Use-case UC12.

2.2 Functional Requirements

2.2.1 Content Management

Following are the requirements for Content Management:

1. The system shall allow content creation and posting in various forms from text to video, images, links to articles, as well as links to blogs.
2. The system shall be able to allow users to give reaction on content that is posted through likes or comments.
3. The system shall enable users to share content posted on the platform with their family & friends.
4. The system shall allow users to tag their friends on any post.

5. The system shall send the reported content to 13 random users to verify whether it is explicit or not.

2.2.2 Profile Management

Following are the requirements for Profile Management:

1. The system shall allow users to make changes in their personal information, including profile or cover photos.
2. The system shall enable users to manage and customize their timeline.
3. The system shall allow users to change their passwords.
4. The system shall enable users to manage their visibility setting for posts.
5. The system shall enable users to customize notifications preferences.
6. The system shall implement KYC-mechanism to cater the boots.

2.2.3 Monetization and Rewards

Following are the requirements for Monetization and Rewards:

1. The system shall enable content creators to monetize their content without requiring any premium version.
2. Upon registration, the system shall give a reward token to new users to enable the transactional process.
3. The reward system shall enable rewards based on the user's choice to view advertisements.

2.2.4 Community Management & Interactions

Following are the requirements for Community Management & Interactions:

1. It shall give the opportunity to form groups with family and friends, sharing private interactions.
2. The system shall support forum creation based on the subject and notify participants when an event has occurred.
3. The system will offer messaging functionality where users will have direct private chats with their friends.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are some examples of documenting guidelines.

2.3.1 Reliability

1. The system shall have 99.9% availability, which doesn't include any scheduled maintenance.
2. The system shall ensure backup and recovery mechanisms in case of any failure to restore user's data.

2.3.2 Usability

1. The interface shall be intuitive and easy to use by people of any age.
2. The system shall include search functionality that enables at least an 80% chance of easily locating a friend, content, or group.
3. The system shall provide the users with preference options to display the content, for instance, newest or most popular. These shall be instantaneous selections.

2.3.3 Performance

1. The system shall be able complete posting and content creation operations with in 5 seconds under normal circumstances.
2. The system shall ensure that at least 1000 users are supported simultaneously with out any decline in performance.

2.3.4 Security

1. The system shall ensure that user passwords and sensitive data are encrypted both in transit and at rest.
2. The system shall have a role-based access control that will restrict actions according to user roles.

2.4 Domain Model

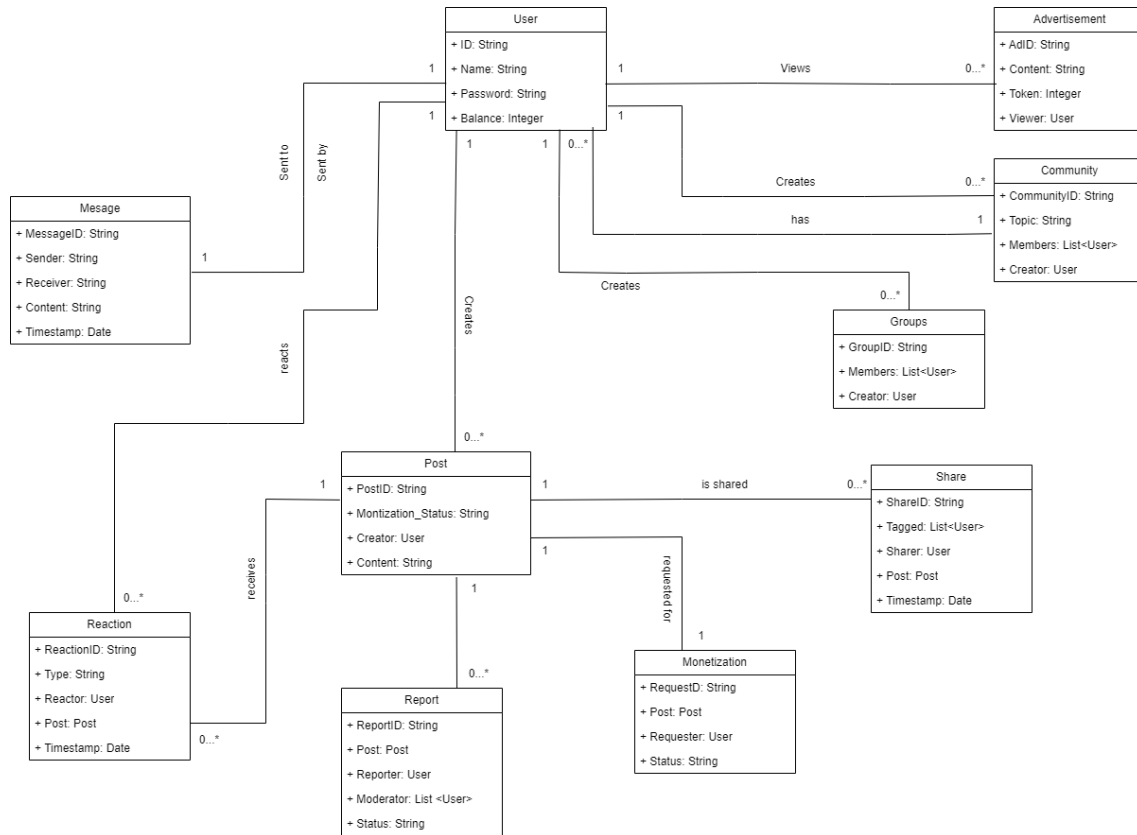


Figure 2.2: Domain Model Diagram.

Chapter 3

System Overview

It is a blockchain-based social media system that enhances the interaction of users while safeguarding the data confidentiality and ownership rights over the contents. It allows users to create, share, and sell the content in a decentralized manner.

The platform benefits from concerns regarding the private data handling of traditional social media and generates revenue without any intermediaries by permitting the users to monetize their contents, in a community-driven manner.

3.1 Architectural Design

3.1.1 Modular Program Structure

The system adopts modular architecture with components that are independently functional but maintain interrelationship. It ensures scalability and maintainability and integrability of blockchain functionalities.

3.1.1.1 Content Management

- **Functionality:** Allow users to generate content and share it while giving ownership right through blockchain.
- **Relationship:** Coordinates with Profile Management in linking up the content to the user profiles and identity validation for permission management.

3.1.1.2 Profile Management

- **Functionality:** It enables the user to manage his information and account settings, such as privacy controls, through a decentralized identity system.
- **Relationship:** Works cooperatively with Monetization and Rewards to track the usage and provide transparent monetization.

3.1.1.3 Monetization and Rewards

- **Functionality:** Enables users to monetize content in tokens and earn rewards for their activities through smart contracts.
- **Relationship:** It aligns with Community Management to encourage user involvement and participation.

3.1.1.4 Community Management & Interactions

- **Functionality:** Enabling groups, event notifications, and private messaging with the ability to ensure protection via blockchain-based encryption.
- **Relationship:** Collaborates with Content Management to facilitate and promote sharing across groups, thereby allowing users to collaborate.

3.1.2 Architecture Diagram

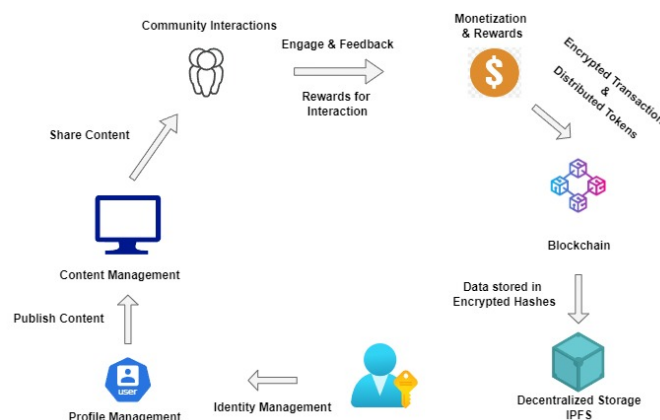


Figure 3.1: Architecture Diagram

3.2 Design Models

3.2.1 Activity Diagram

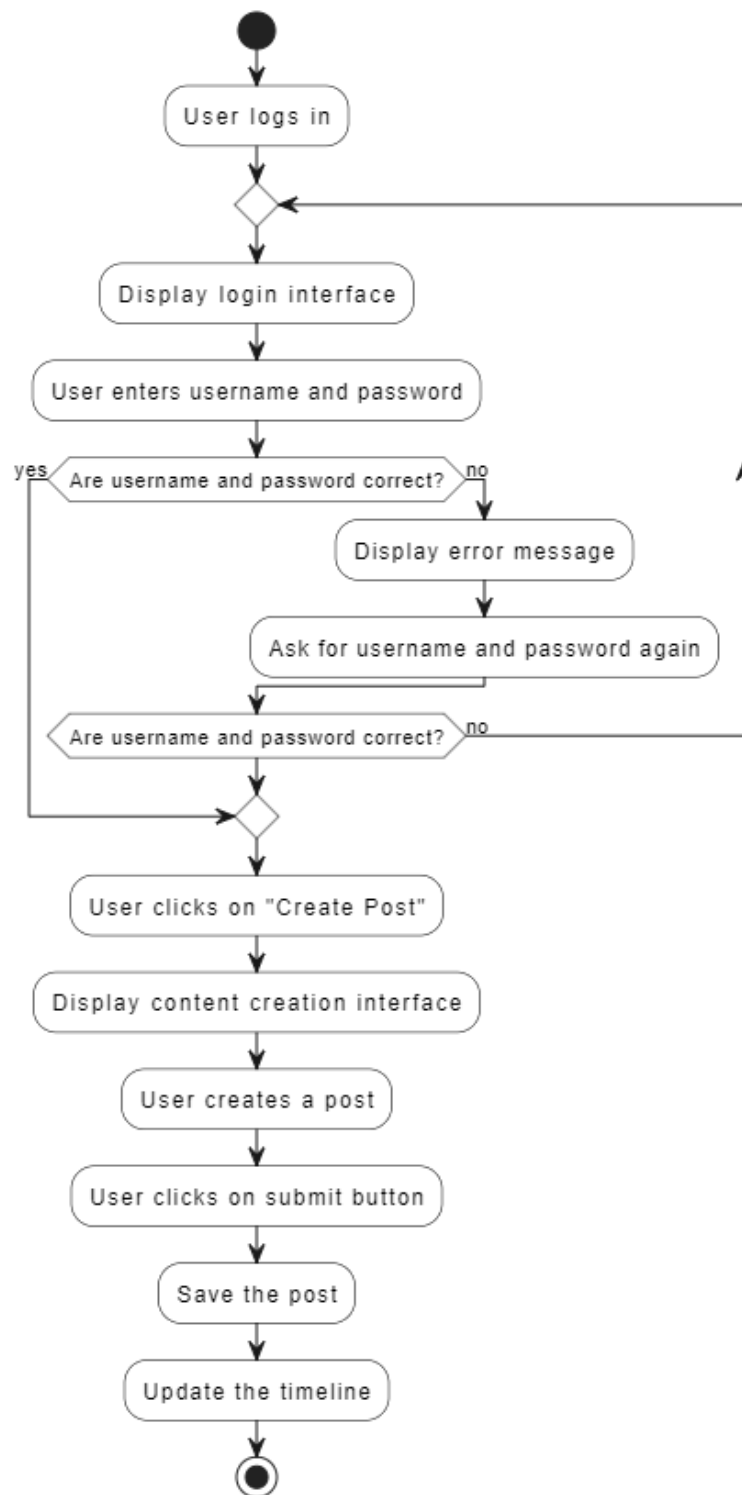


Figure 3.2: Activity Diagram UC1: Content Creation.

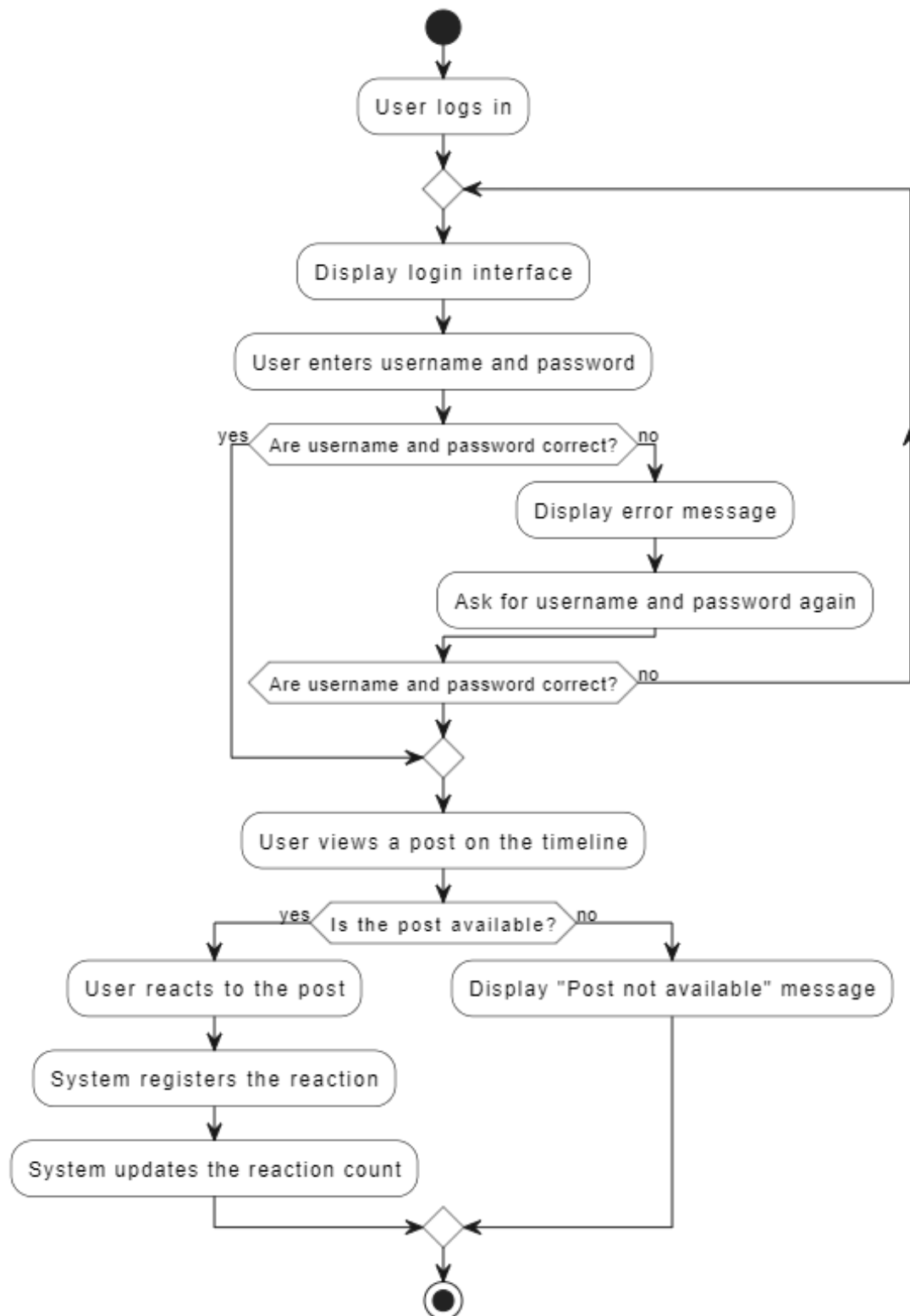


Figure 3.3: Activity Diagram UC2: Content Reaction.

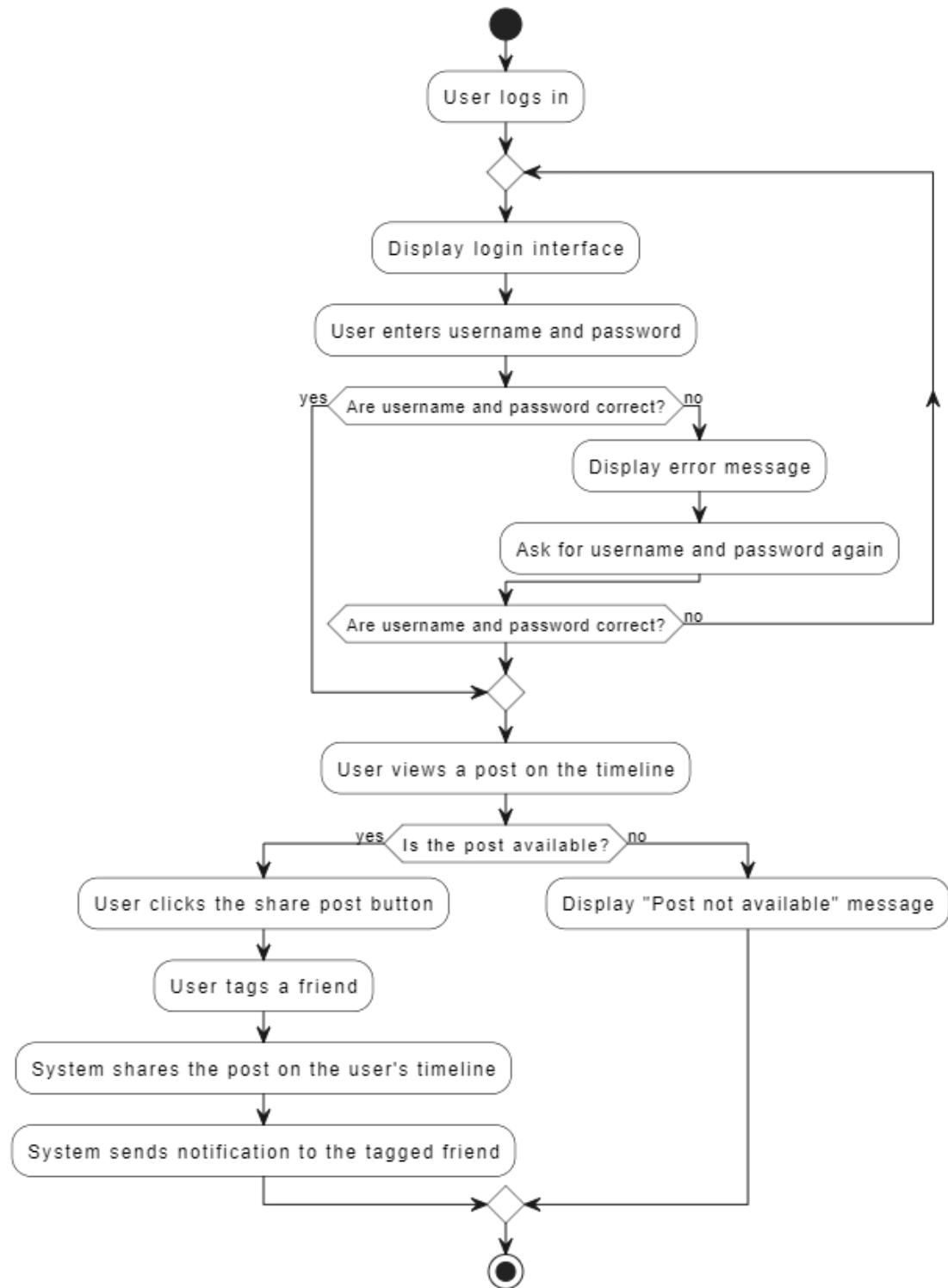


Figure 3.4: Activity Diagram UC3: Content Sharing.

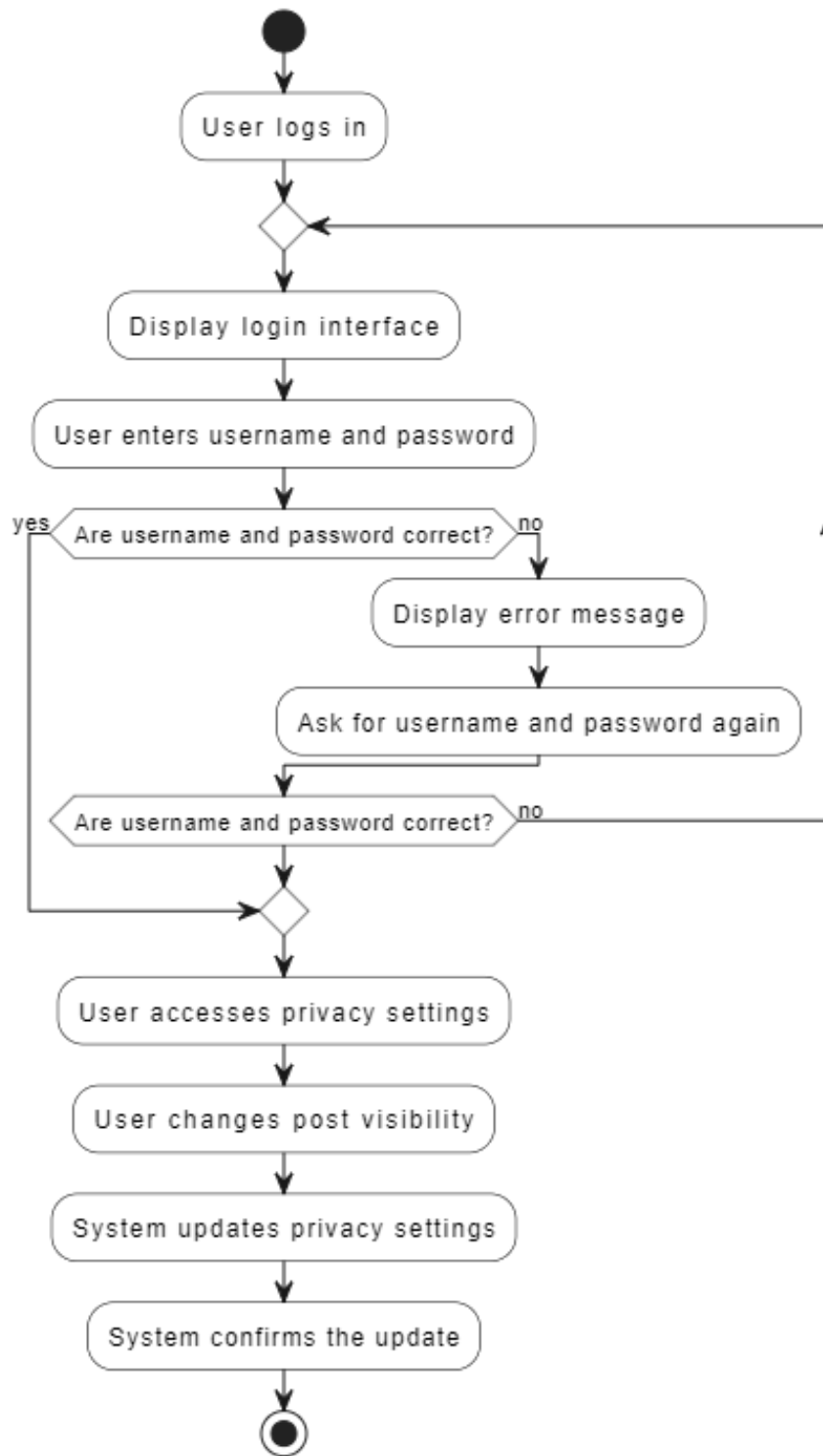


Figure 3.5: Activity Diagram UC4: Privacy Settings Management.

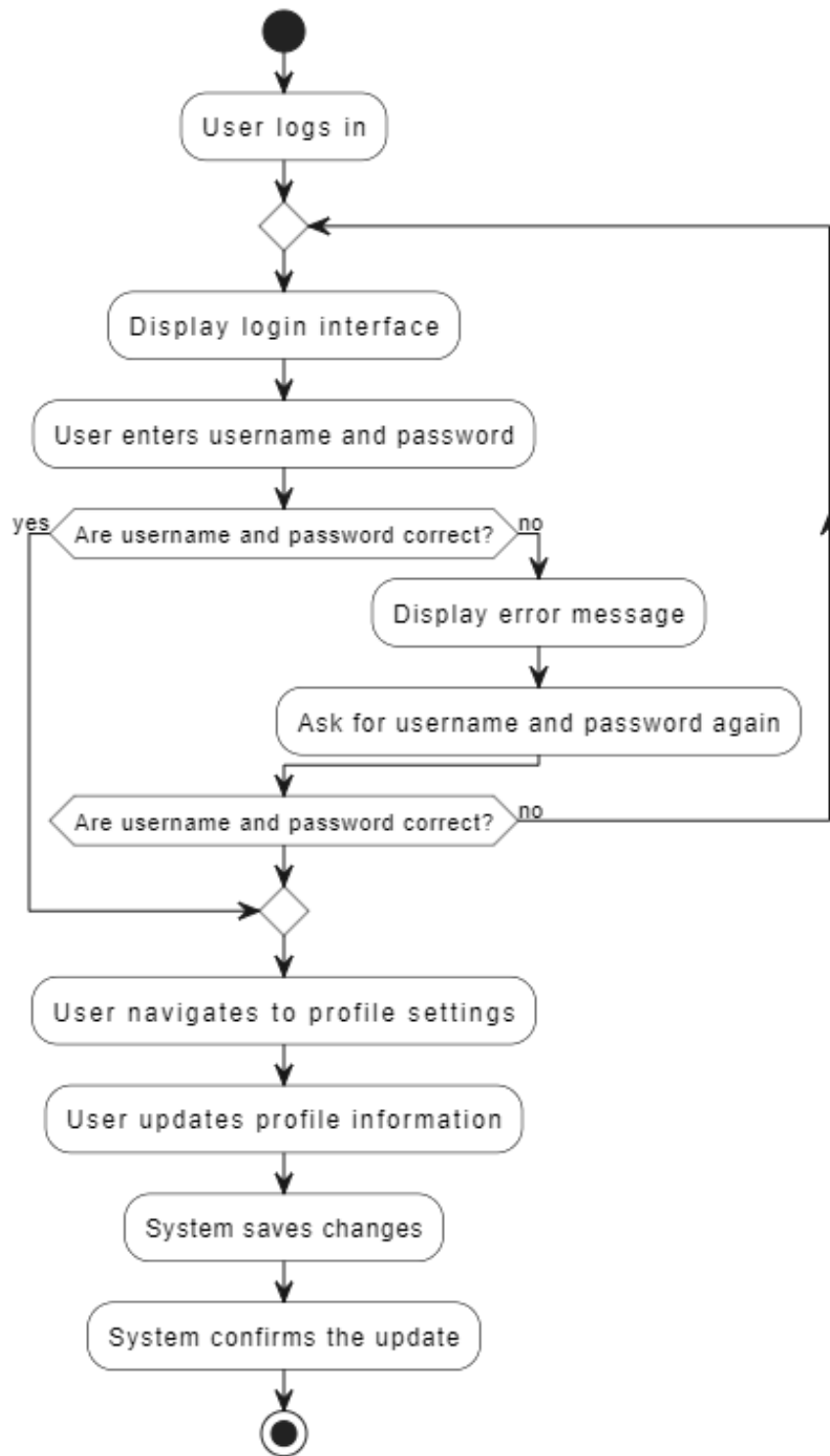


Figure 3.6: Activity Diagram UC5: Profile Management.

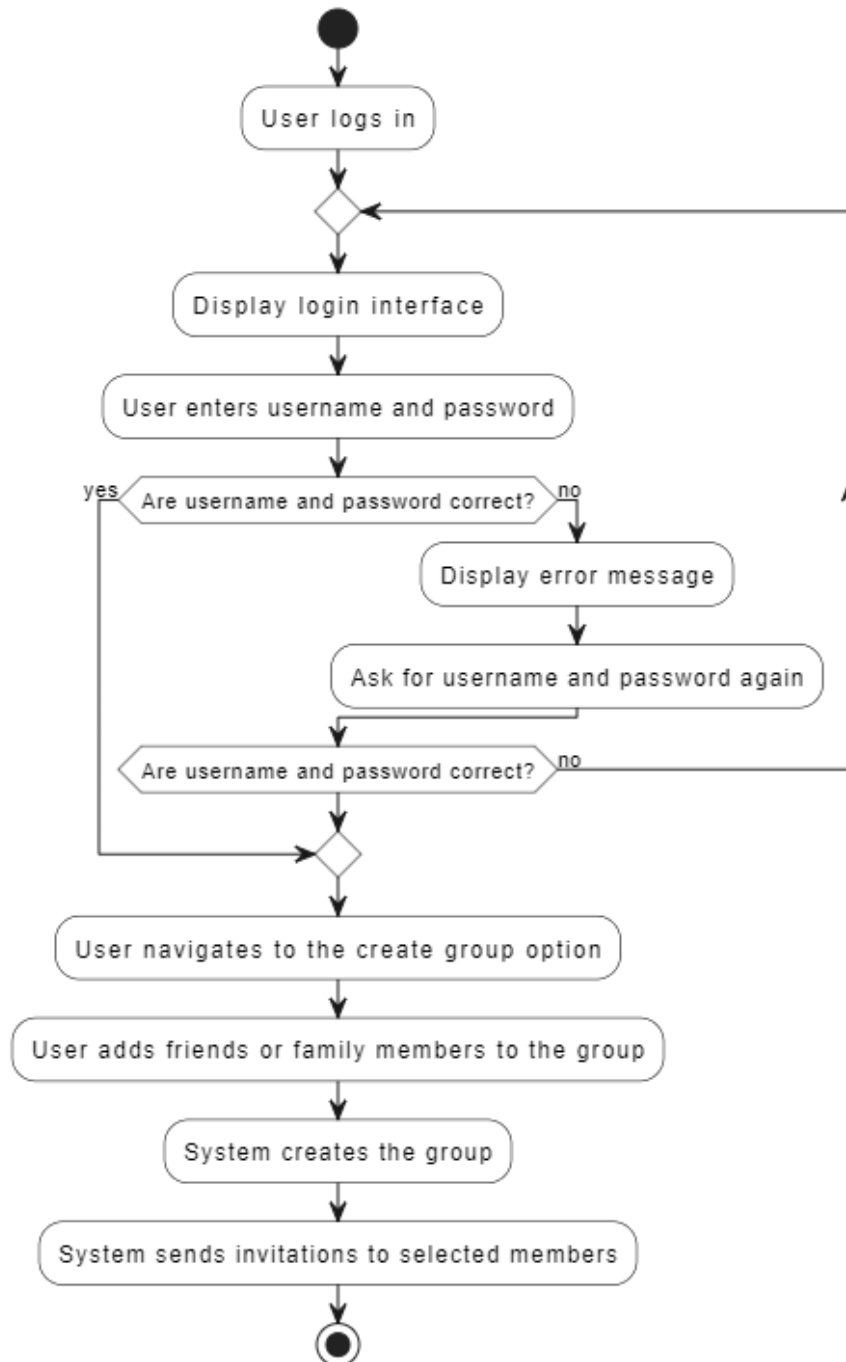


Figure 3.7: Activity Diagram UC6: Group Creation.

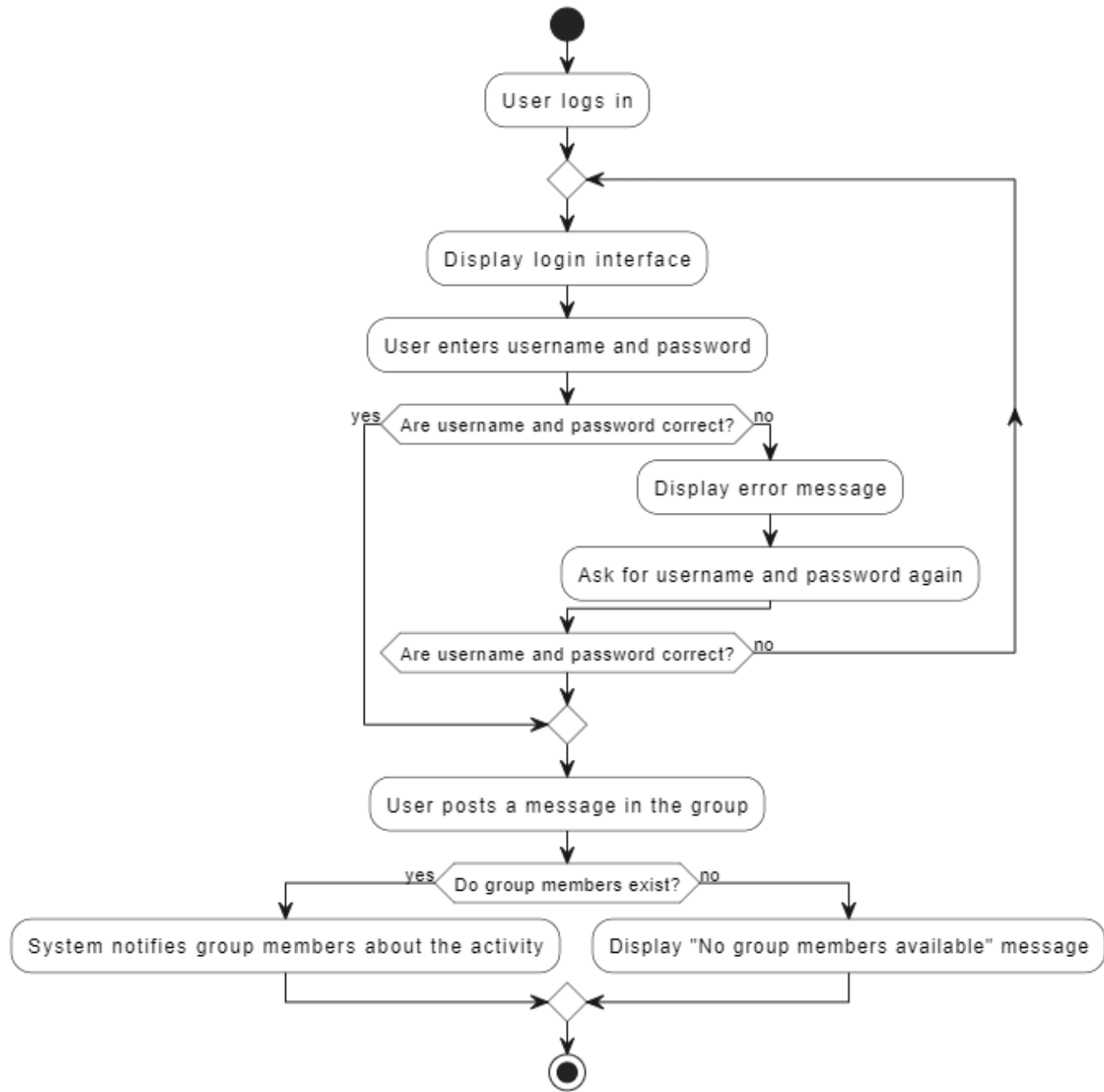


Figure 3.8: Activity Diagram UC7: Group Interaction.

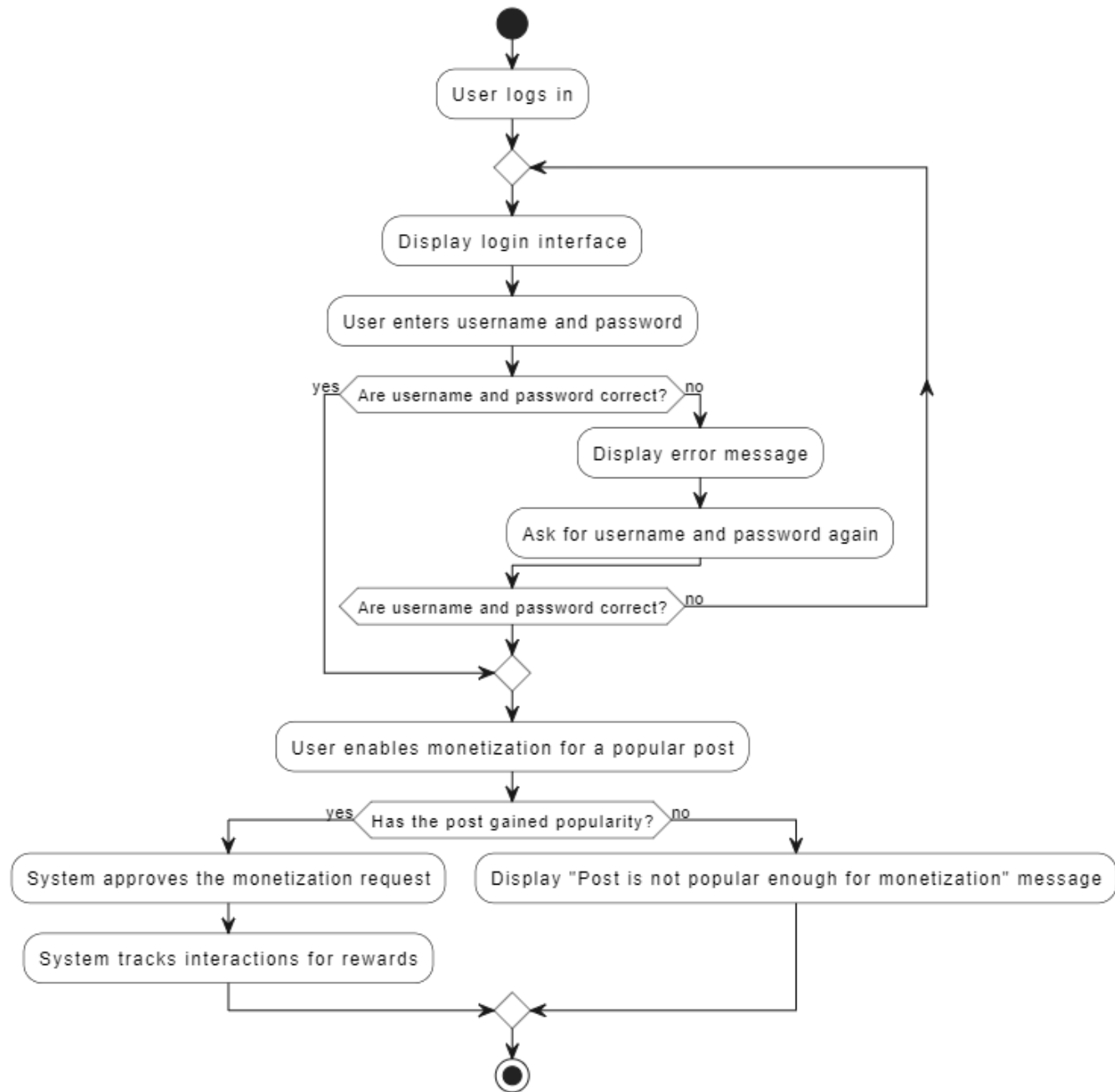


Figure 3.9: Activity Diagram UC8: Content Monetization.

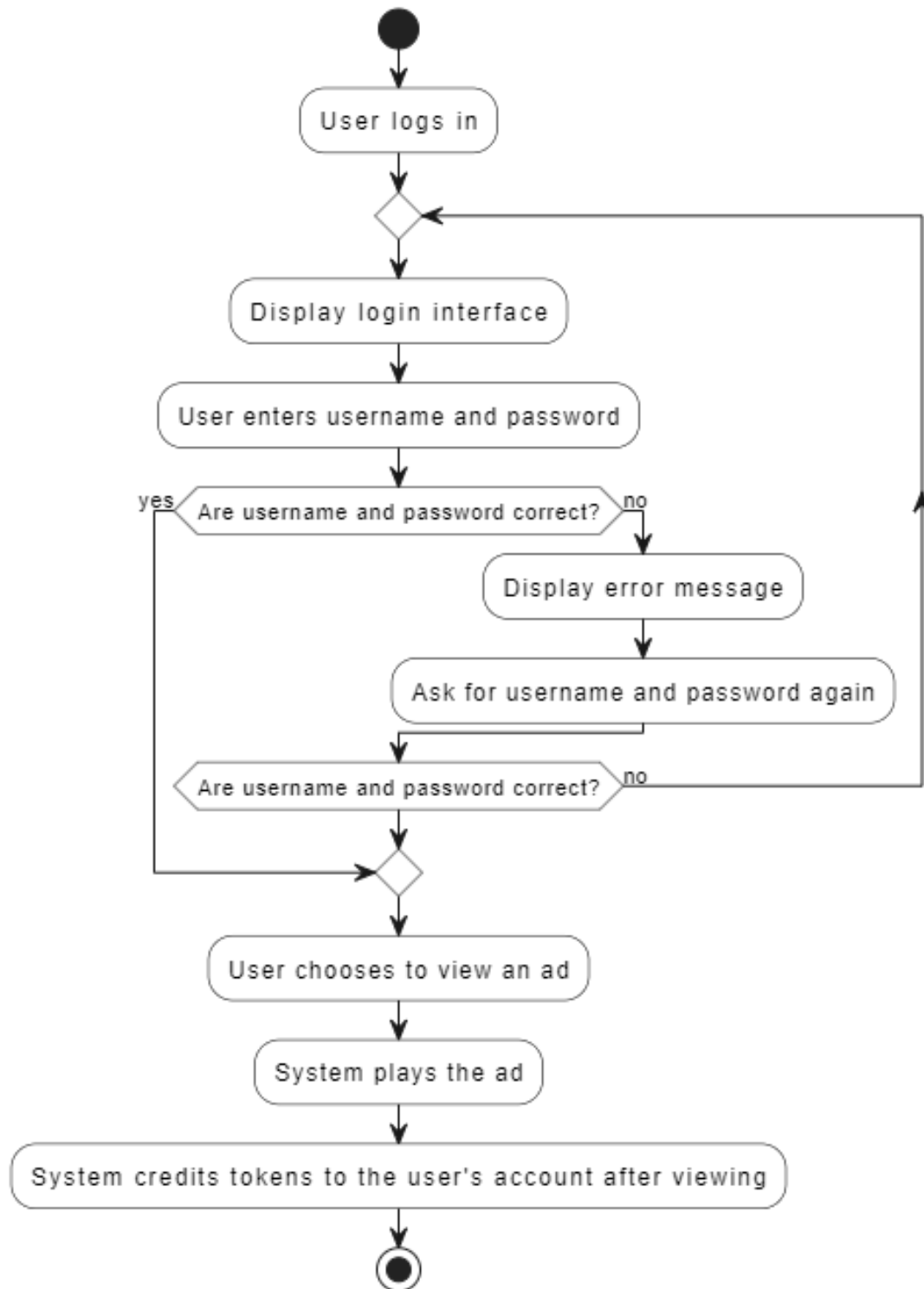


Figure 3.10: Activity Diagram UC9: View ads for rewards.

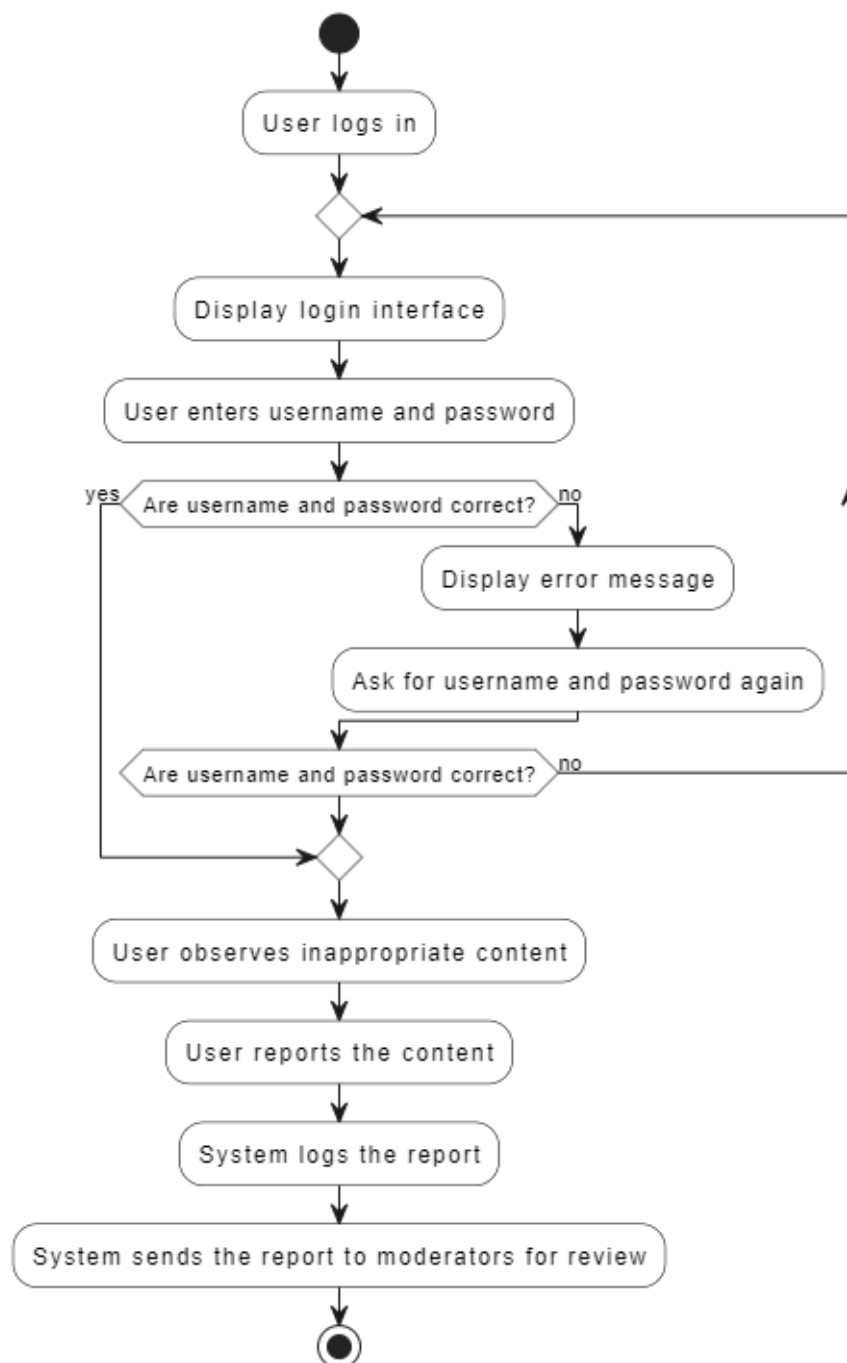


Figure 3.11: Activity Diagram UC10: Report Content.

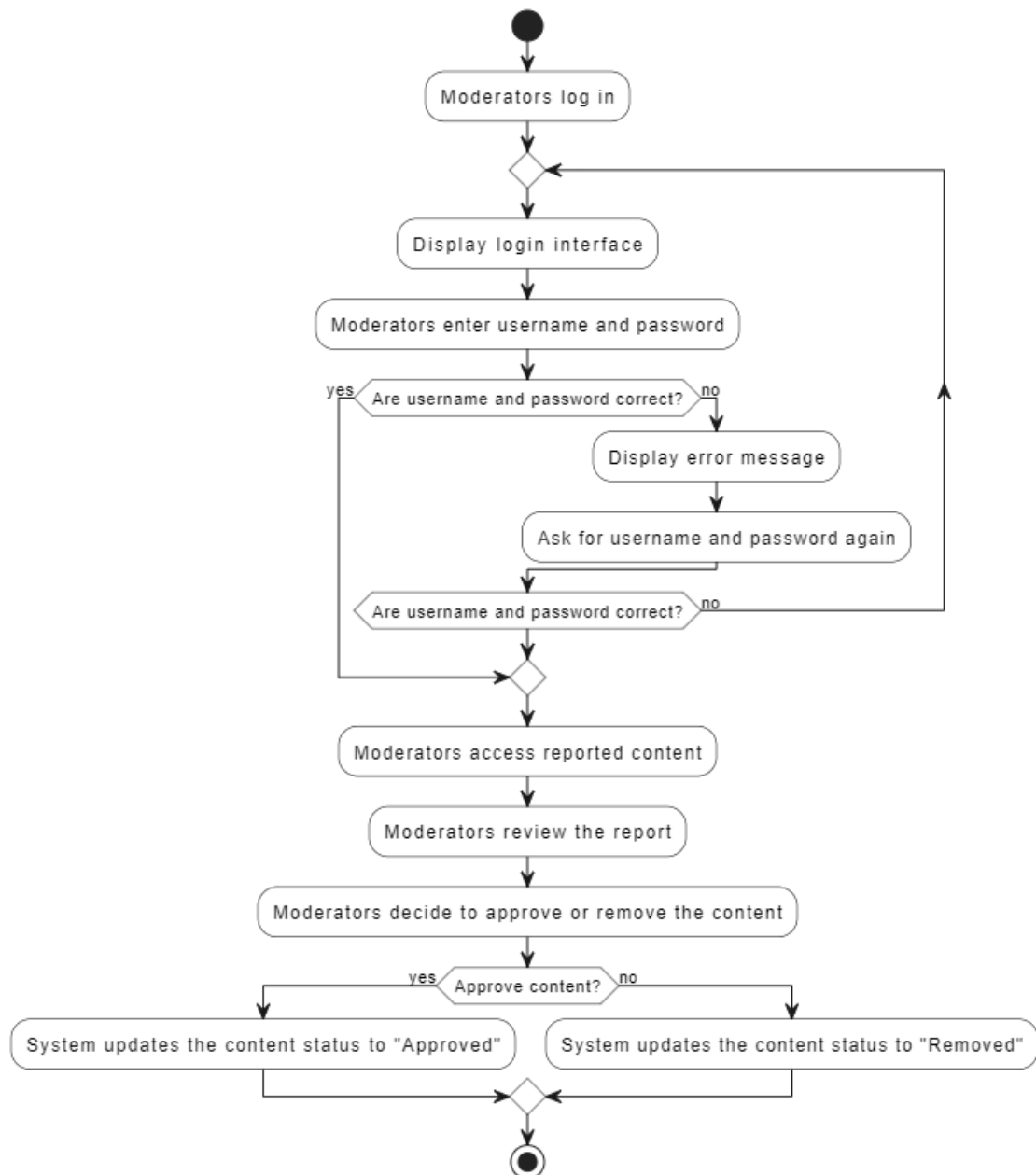


Figure 3.12: Activity Diagram UC11: Review Reported Content: Private Messaging.

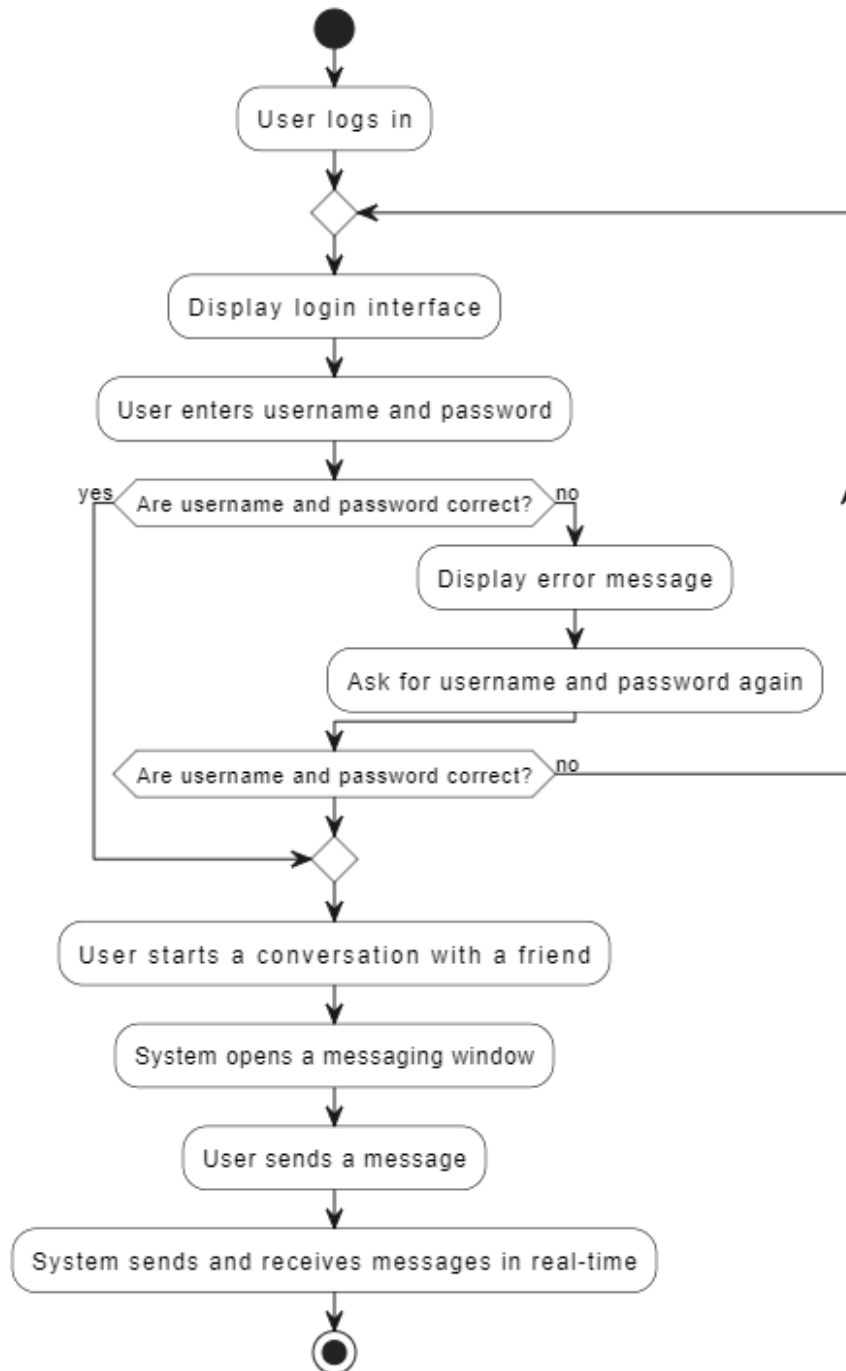


Figure 3.13: Activity Diagram UC12: Private Messaging.

3.2.2 Class Diagram

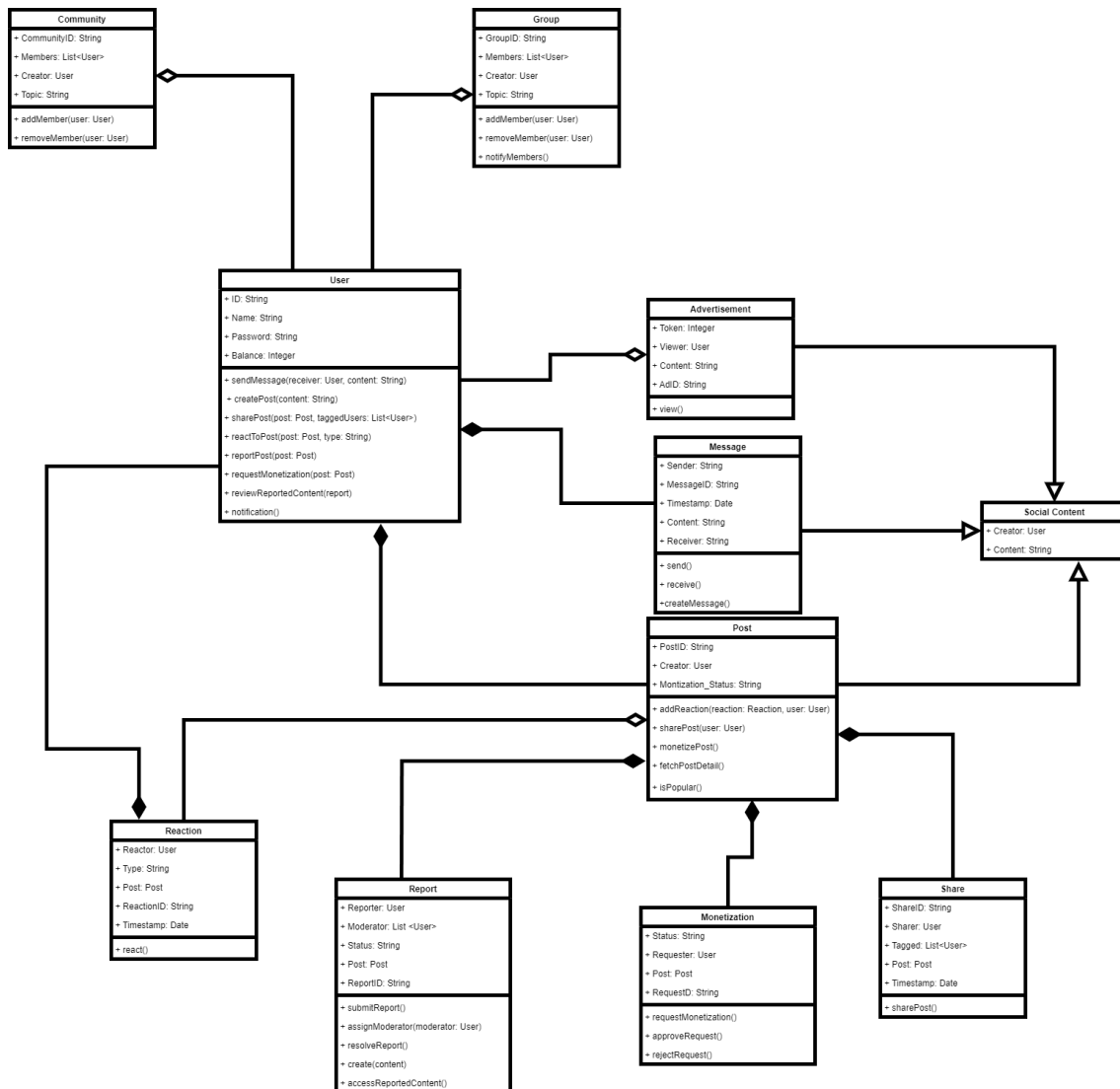


Figure 3.14: Class Diagram.

3.2.3 Class-level Sequence Diagram

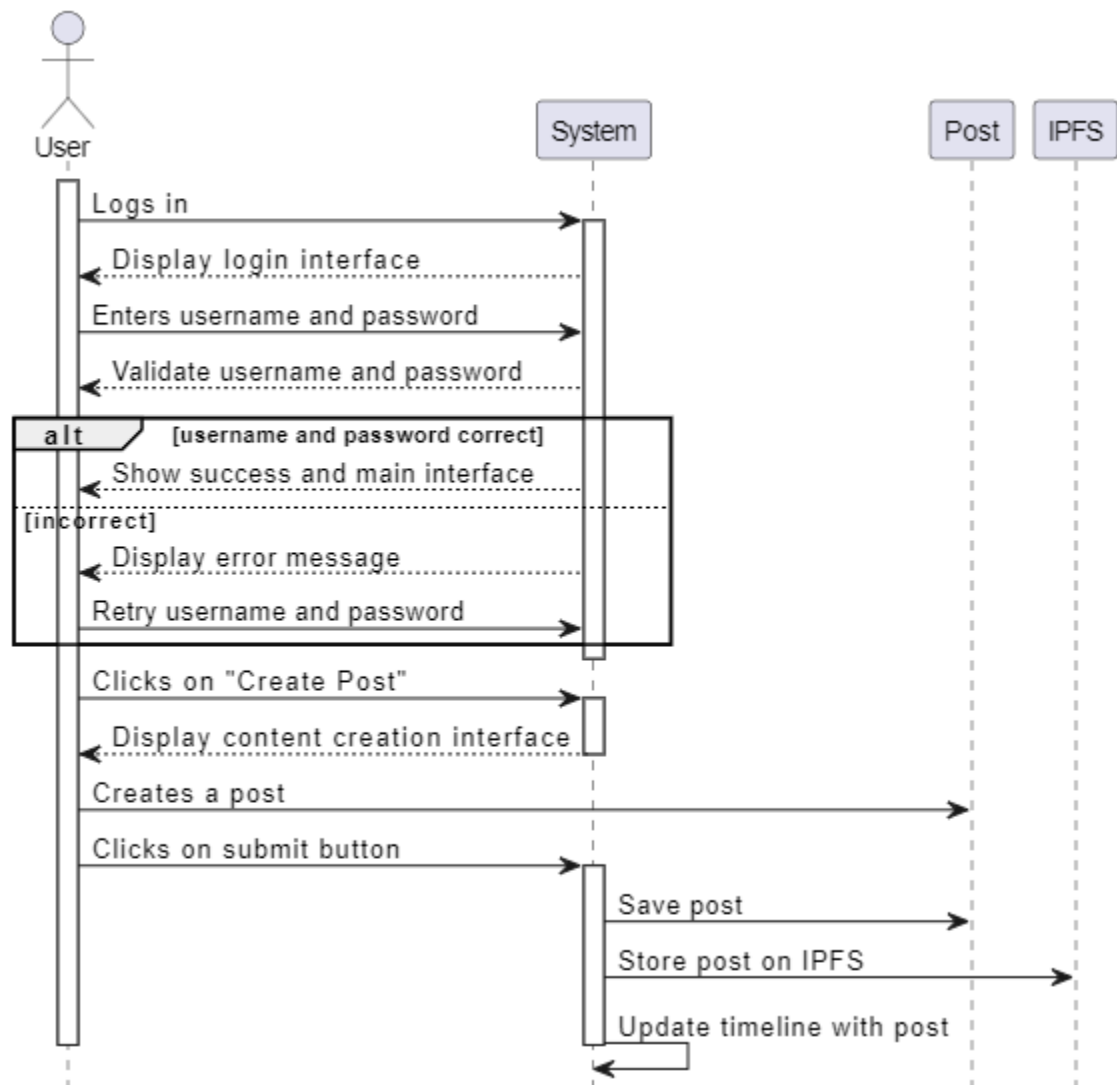


Figure 3.15: Sequence Diagram UC1: Content Creation.

3. System Overview

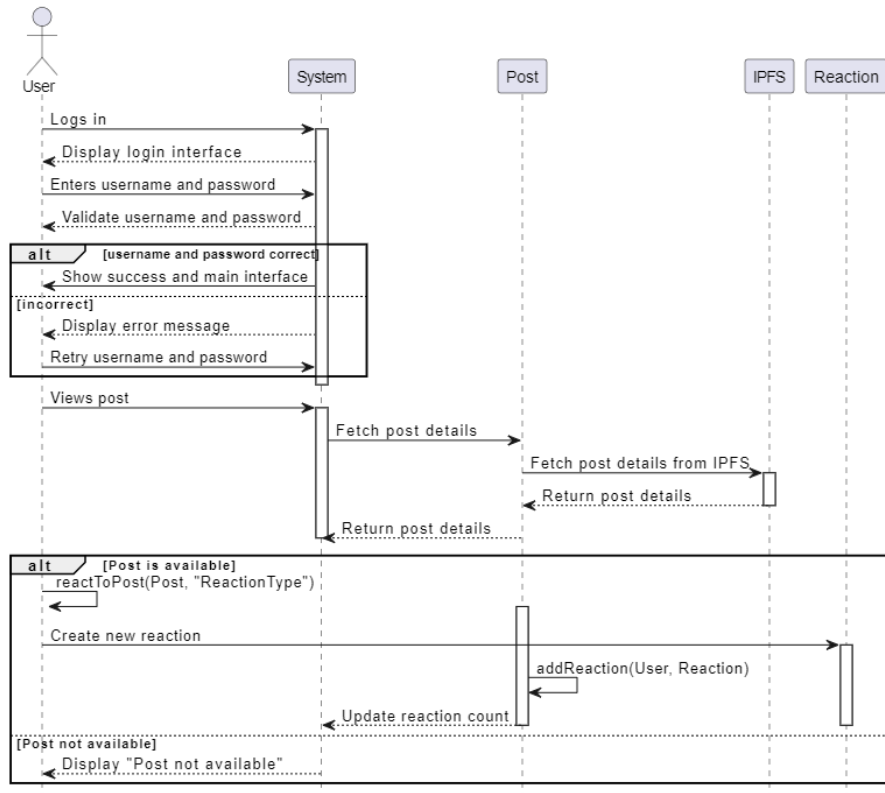


Figure 3.16: Sequence Diagram UC2: Content Sharing.

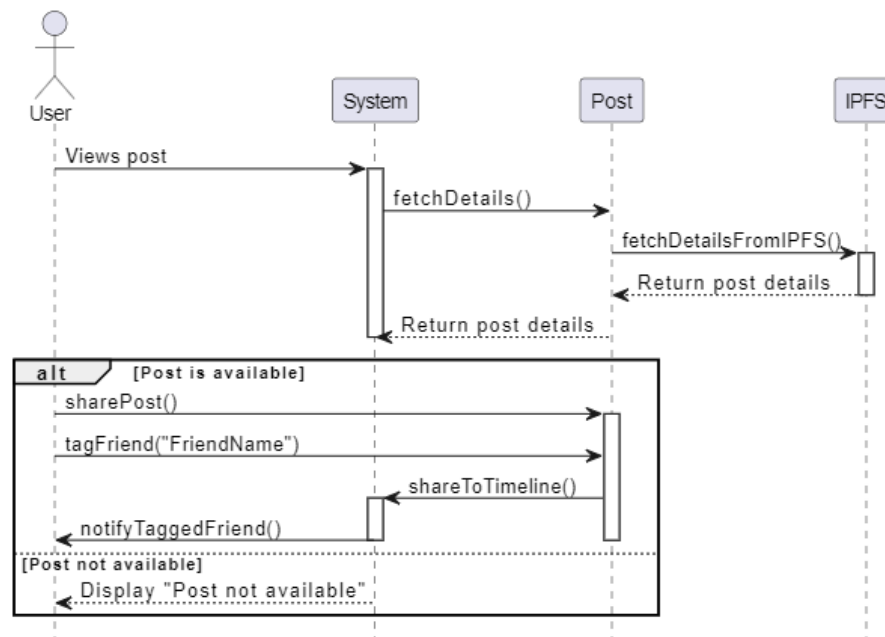


Figure 3.17: Sequence Diagram UC3: Content Reaction.

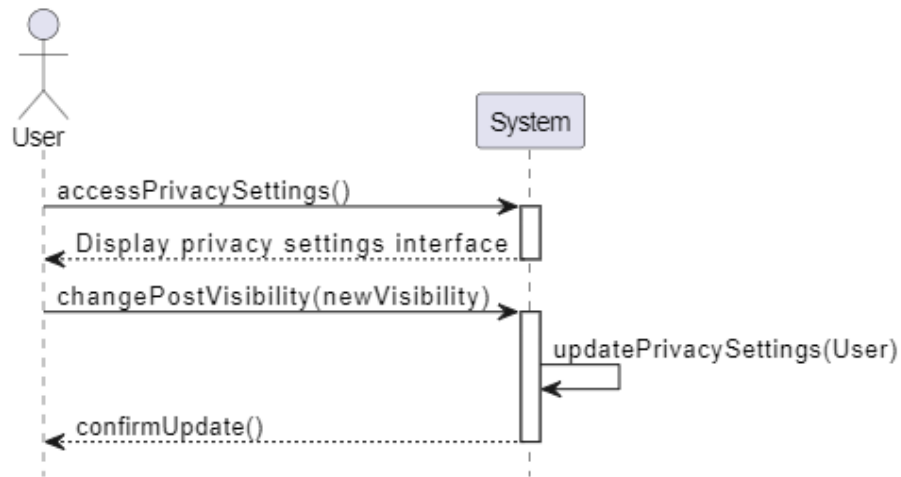


Figure 3.18: Sequence Diagram UC4: Privacy Settings Management.

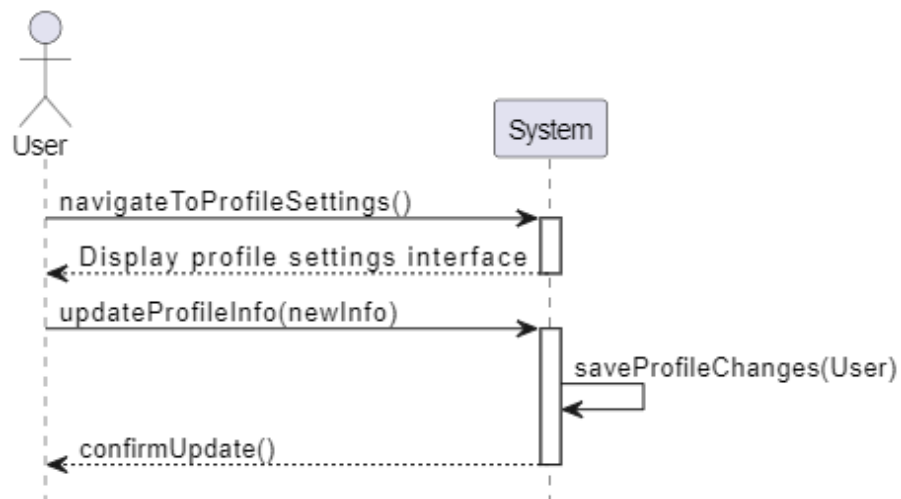


Figure 3.19: Sequence Diagram UC5: Profile Management.

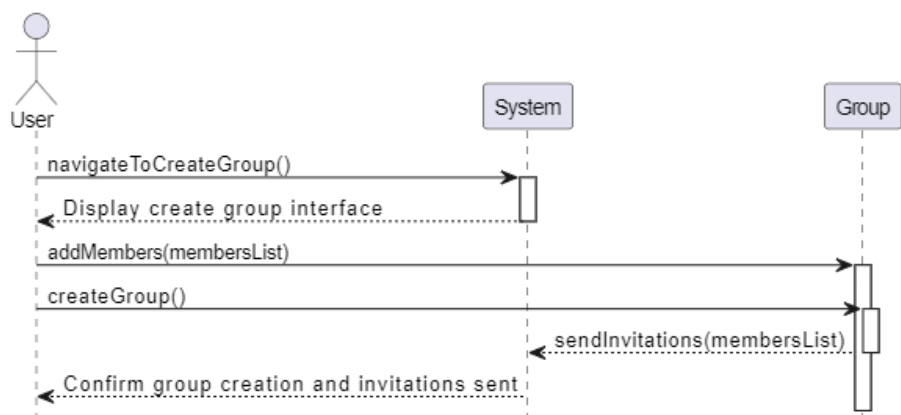


Figure 3.20: Sequence Diagram UC6: Group Creation.

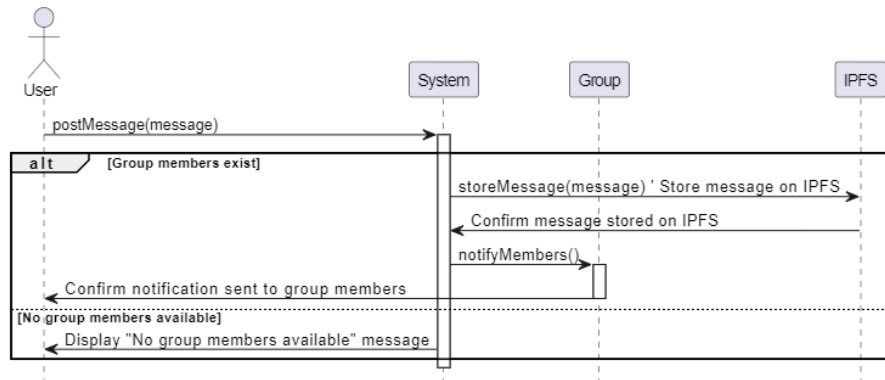


Figure 3.21: Sequence Diagram UC7: Group Interaction.

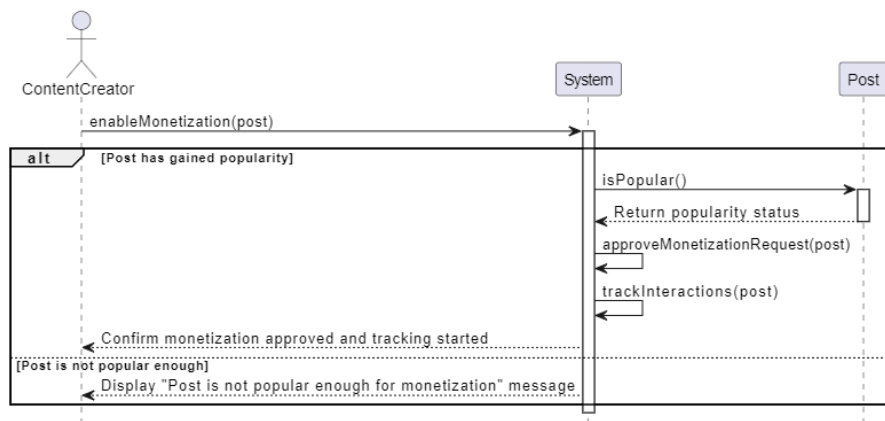


Figure 3.22: Sequence Diagram UC8: Content Monetization.

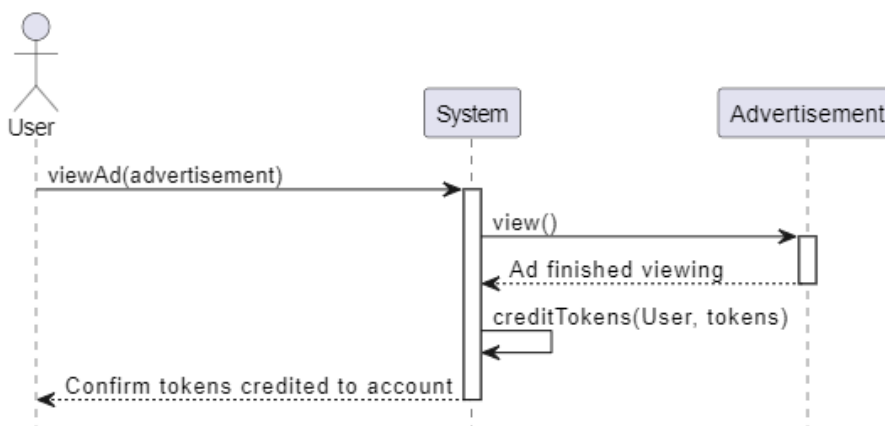


Figure 3.23: Sequence Diagram UC9: View ads for rewards.

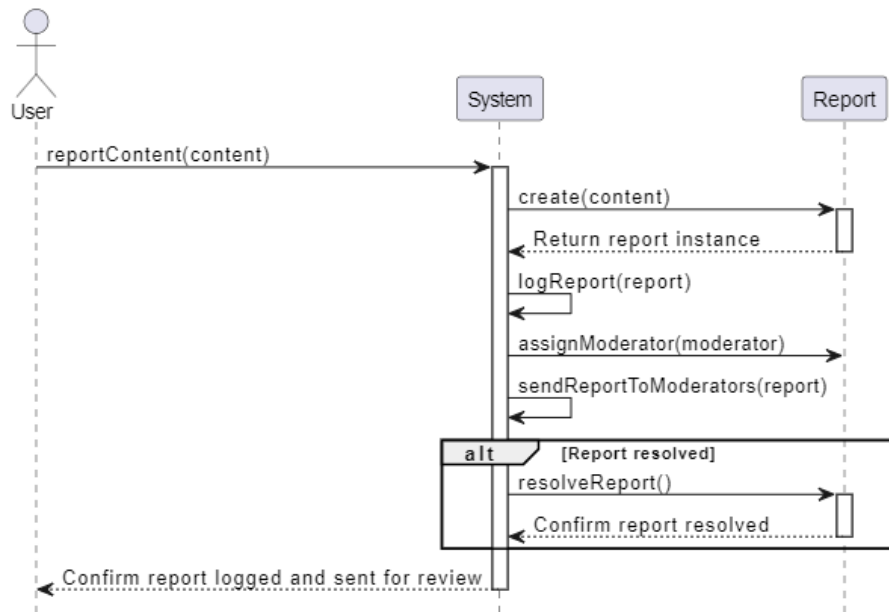


Figure 3.24: Sequence Diagram UC10: Report Content.

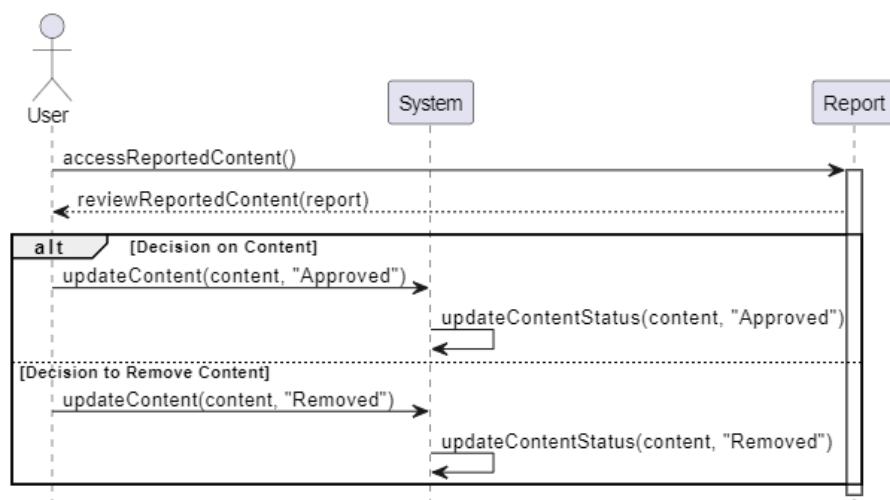


Figure 3.25: Sequence Diagram UC11: Review Reported Content.

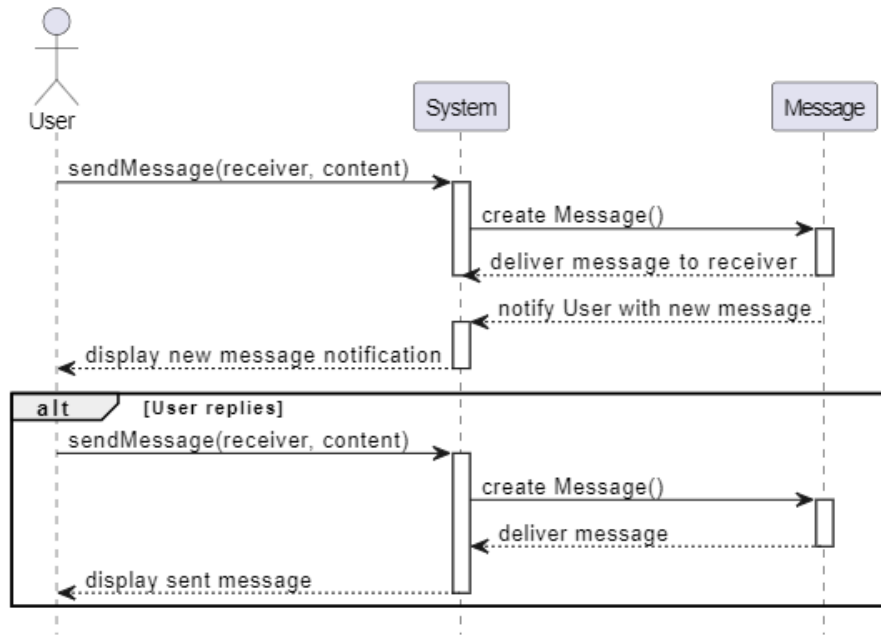


Figure 3.26: Sequence Diagram UC12: Private Messaging.

3.3 Data Design

3.3.1 Key Entities

- **Users:** The basic information such as ID,name,balance will be placed in this table. Password will be hashed and the content on profile will be stored on IPFS as storage is decentralized.
- **Posts:** Posts are the key aspect in content creation. The posts made by users such as videos,text will be stored on IPFS, with all required fields.
- **Reactions:** The given by users are key aspect for interaction between users. They can be stored on chain with reference to user and post on the blockchain.
- **Message:** Chats between users can be encrypted and stored on IPFS, with fields such as sender,receiver etc.. on the blockchain.
- **Groups:** Decentralized management of group and community membership, which includes group identifiers and users, allows tracking of membership on-chain with the possibility of storing off-chain group data, such as topics, on IPFS.
- **Advertisements:** Ad's that are posted can be referenced with tokens and store their content on IPFS.

- **Report:** User's can report content or users and this information can be stored on IPFS along with moderators who evaluate the post.
- **Monetization:** The request for monetization can be made by users to request for tokens rewarded and are also stored with approval or rejection by smart contracts on the blockchain.
- **Share:** A record of users posting will be kept on-chain and the original content will be stored on IPFS.

3.3.2 Data Storage Items

3.3.2.1 IPFS

The major content that includes videos, text, images, etc.. will be stored using IPFS mechanism. Whenever a content is posted by user it is pinned by IPFS and in the database a CID is referenced.

3.3.2.2 Smart Contracts

In this blockchain-based application, smart contracts will manage various functionalities: user accounts for balance tracking, profile changes, and password recovery; content management for post creation, sharing, and reactions, with metadata stored on-chain and content referenced on IPFS; monetization requests for validating and distributing token rewards; group management for adding or removing users from communities; and ad management for overseeing ad tokens and tracking interactions with advertisements.

3.3.2.3 Blockchain Integration

Every action, including posting content, reacting, or sending messages, is written to the blockchain and provides absolute transparency and immutability. This leaves the blockchain free to manage tokens, user balances, and monetization rewards, while IPFS manages all the actual data.

3.3.3 IPFS Data-Flow

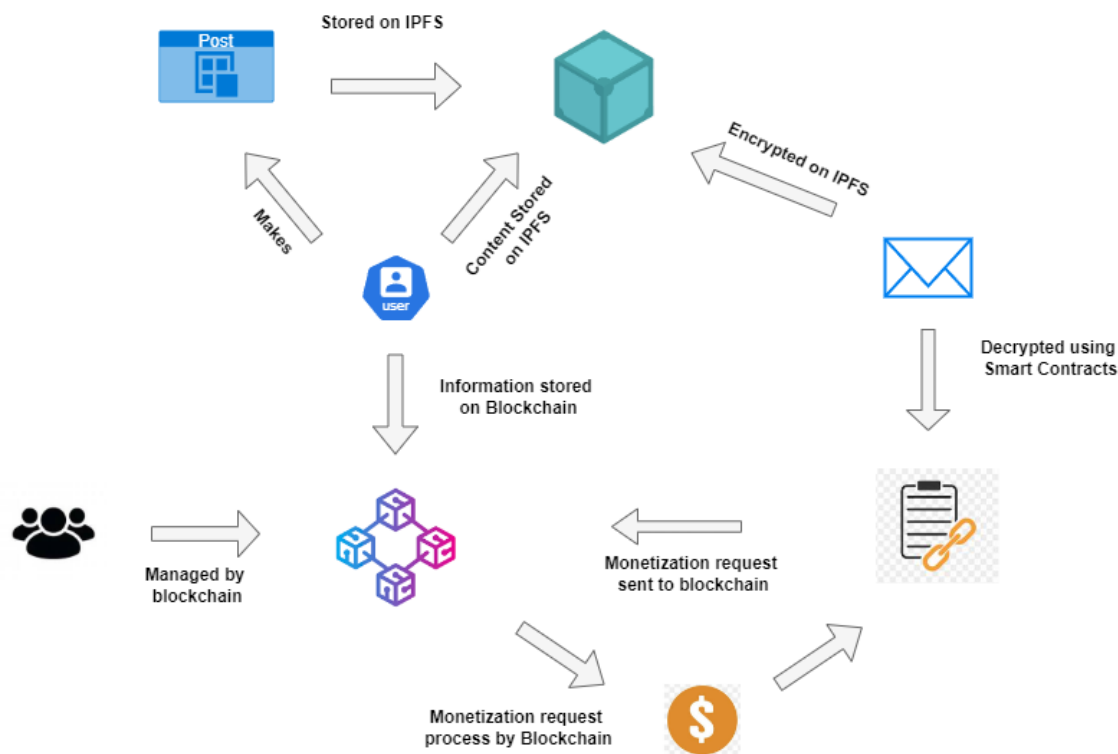


Figure 3.27: IPFS Data Flow

3.3.4 Merkle Tree

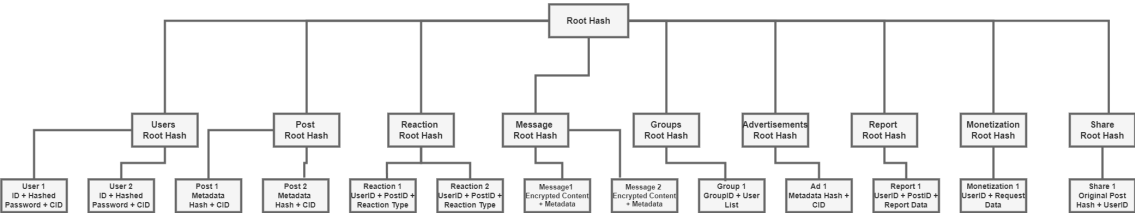


Figure 3.28: Merkle Tree

Chapter 4

Conclusions and Future Work

In conclusions, the development a decentralized social media platform is make a step towards a revolutionize the way of interaction between people. By leveraging blockchain technology, it will provide users a decentralized platform where users can express their thoughts freely without fear of censoring their content by a third party or centralized authority. This will encourage users to take ownership of the content that is shared by them and they will also be able to earn rewards through monetizing their content. Their data will be protected and will not be breached by other people. Our goal is to create a environment for users where their right to freedom of speech is protected and they can express their feelings in front of others.

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